

JERSEY AIRPORT SCHEDULE DASHBOARD

Overview

This dashboard was developed using Python Dash, Plotly Express, and dash-mantine-components and is designed to help airport operations staff understand and analyse a flight schedule. In particular, the application focuses on providing interactive visualisations of flight volume analysis and capacity analysis.

The application uses a FastAPI backend for data processing and analytics and a dark theme with neon accent colours is used throughout to improve readability and create a modern operations-dashboard appearance.

Design Decisions

The application uses a simple navigation bar to allow users to switch between three different pages: Overview, Flight Volume, and Capacity.

The overview page is comprised of two main sections: Key Metrics and Flight Volume by hour. Key metrics currently included are total flights, top airline, and top destination for the selected date and this section could be expanded to include further metrics or a summary table of flights that day. A line graph is used to provide an immediate understanding of when activity peaks occur throughout the day.

The Flight Volume page focuses on analysing traffic patterns. Users can filter data by date range, airline, and destination. Visualisations include:

- Bar Chart- Flights per Hour
 - Daily operational peaks
- Line Chart- Flights per Days
 - Shows longer-term schedule trends across the dataset

The capacity analysis page uses the number of seats per flight to visualise total seats per hour across a selected day. This view helps to identify periods where passenger carrying capacity is concentrated and can support decisions relating to terminal resources, staffing levels, security screening demand, and gate utilisation. With further data such as passengers per flight, this page could be improved to provide more accurate estimations of load factors throughout the day.

Assumptions Made

Several assumptions were made while developing the application:

- Departures Only
 - The primary focus was placed on departures as in the sample flight schedule provided, the only departure airport was Jersey Airport.
- Capacity by seats
 - The 'Seats' field was assumed to represent the maximum passenger capacity of each field.
- Airport Codes
 - As the application is designed for use by operations staff, it is assumed that airport names are not required for analysis allowing the use of airport codes for more efficient layout usage and readability.
- Data Quality
 - The application assumes valid flight dates, departure times, seat counts and airport coding, only minimal data cleaning and type conversion were performed.